

Experiment No: 06

Experiment: Construction of Phase diagram for Biphenyl Naphthalene system

Theory:

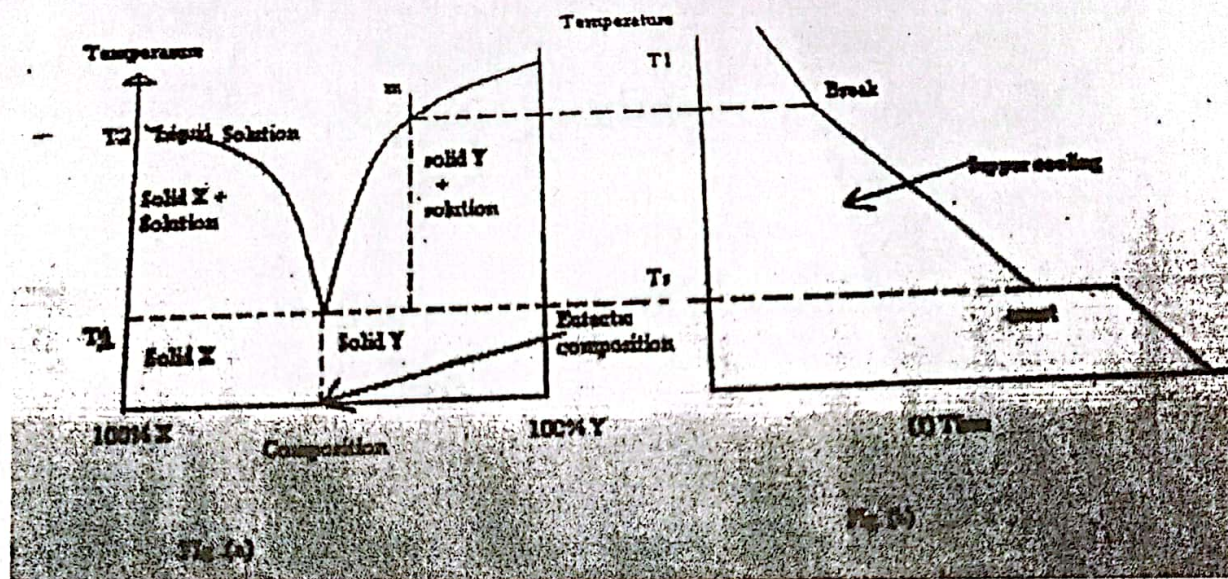


Figure (a)- Phase diagram for a typical binary mixture with simple eutectic formation.

On other principle way of investigating solid-liquid equilibria is by thermal analysis. (i.e.) by the study of the cooling curves (temperature vs time) obtained when the liquid solution is slowly cooled. If a liquid of composition m (fig.(a)) is slowly cooled. The solid Y starts separating out at temperature T . This is accompanied by the evolution of heat (i.e. The negative of the latent heat of fusion). The rate of cooling should therefore, show sharp changes at temperature at which the deposition of solid begins. (temperature T_1) and when it is completed. (i.e. Temperature T_s). In practice, the sharp peaks are often marked by super cooling figure(b). Nevertheless, it is usually possible to locate the break and arrest of the cooling curve, and so to construct a phase diagram.

Procedure:

1. Make mixture of Biphenyl and Naphthalene in which the mole fraction of Naphthalene is roughly 0.16, 0.20, 0.25, 0.33, 0.40, 0.50, 0.67 and 0.83. The total amount of each mixture should be 1g.
2. Place 1 mixture at a time in a test tube and melt by immersing in hot water.
3. Cool the mixture in an air with continuous stirring. Read the temperature as function of time, at half minute intervals. Note also the temperature of first trace of solid separating out. (Up to 10 minutes).
4. Find the melting points of pure Naphthalene and Biphenyl separately.